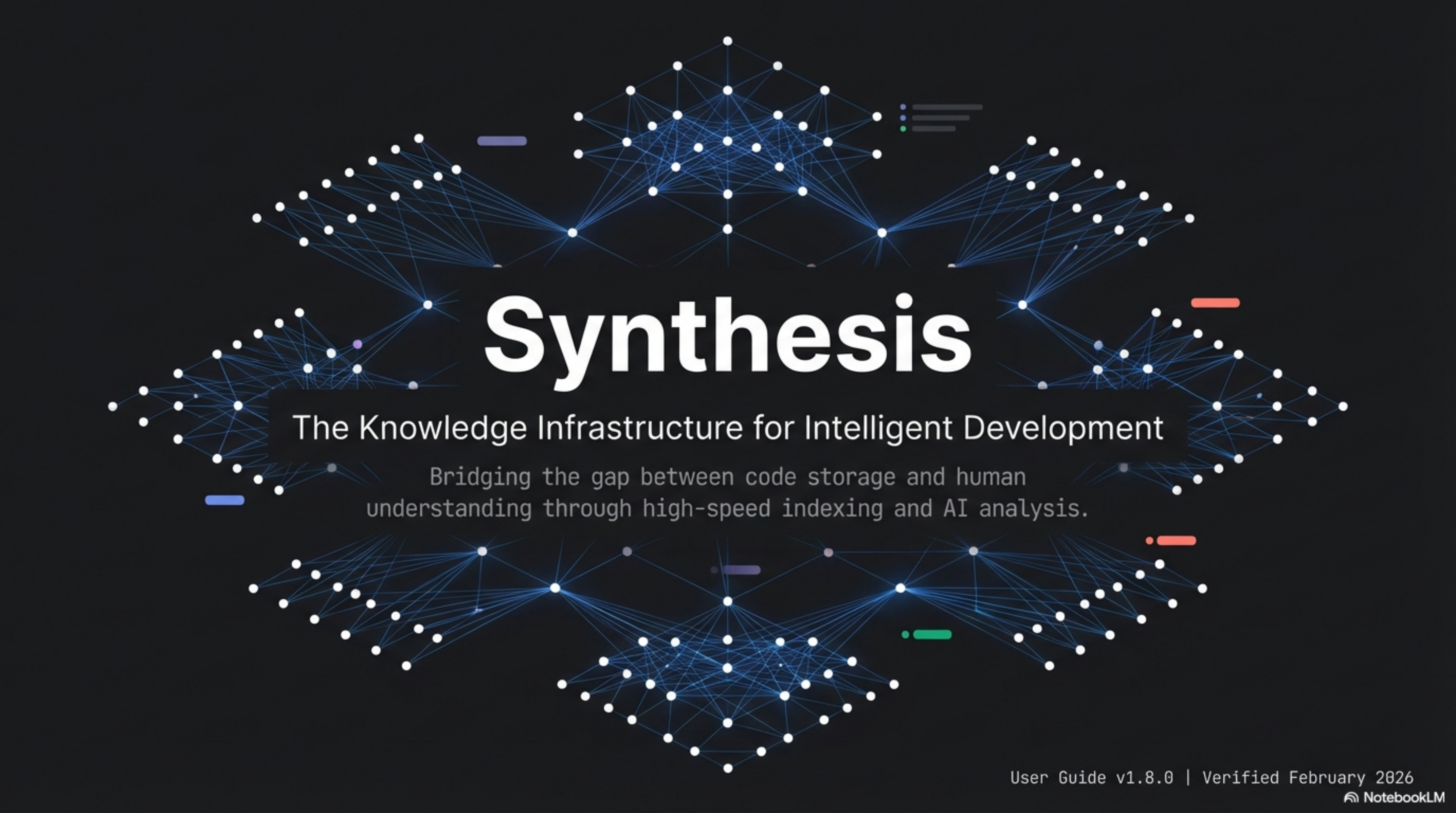


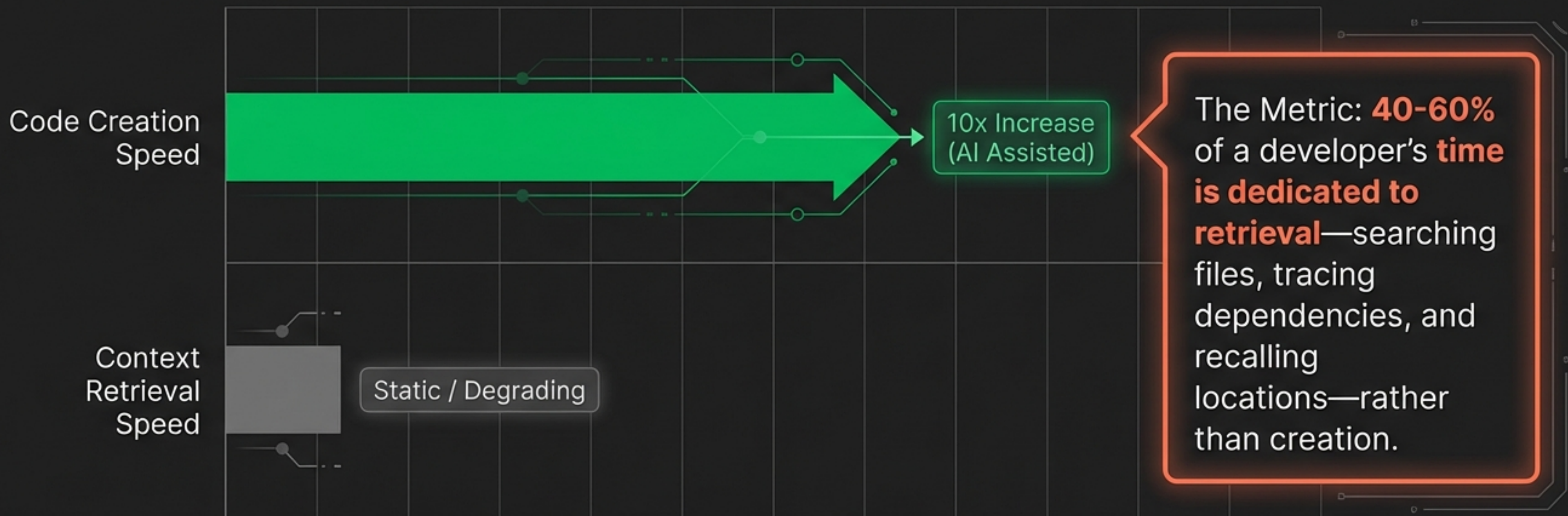


Synthesis

The Knowledge Infrastructure for Intelligent Development

Bridging the gap between code storage and human understanding through high-speed indexing and AI analysis.

The Context Gap in Modern Development



As codebases grow, the human ability to maintain a mental map degrades. The bottleneck is no longer typing syntax; it is locating logic.

Turning Data into Immediate Intelligence



Universal Indexing

Ingests unstructured data (Code, Video, PDF) into a structured graph.

AI Integration

Tracks relationships and provides analysis when configured.

High Efficiency

Storage overhead is only 2-3% of indexed content (e.g., 11.6 MB index for 434 MB content).

The Mechanics of Ingestion

```
$ synthesis scan
> Ingesting workspace...
> Parsing 8,934 files (**/*.java, **/*.py, **/*.md, **/*.pdf)
> Building dependency graph...
[=====] 100%
Scan complete. Index updated.
```

.synthesis/config.yaml

```
includePatterns:
  - '**/*.java'
  - '**/*.py'
  - '**/*.mp4'
excludePatterns:
  - '**/node_modules/**'
  - '**/.git/**'
```

The system crawls the workspace, parses a huge variety of formats, and builds a local relationship graph.

Validated Performance Benchmarks

Data Source: Validated February 2026



200-300

Files Indexed Per Second



< 0.4s

Average Search Speed



< 31s

To Map 429 Cross-Repo
Dependencies

Performance remains stable as complexity scales. Time to first search result for a new project: approx 2-5 minutes.

Perspective: The Developer Experience

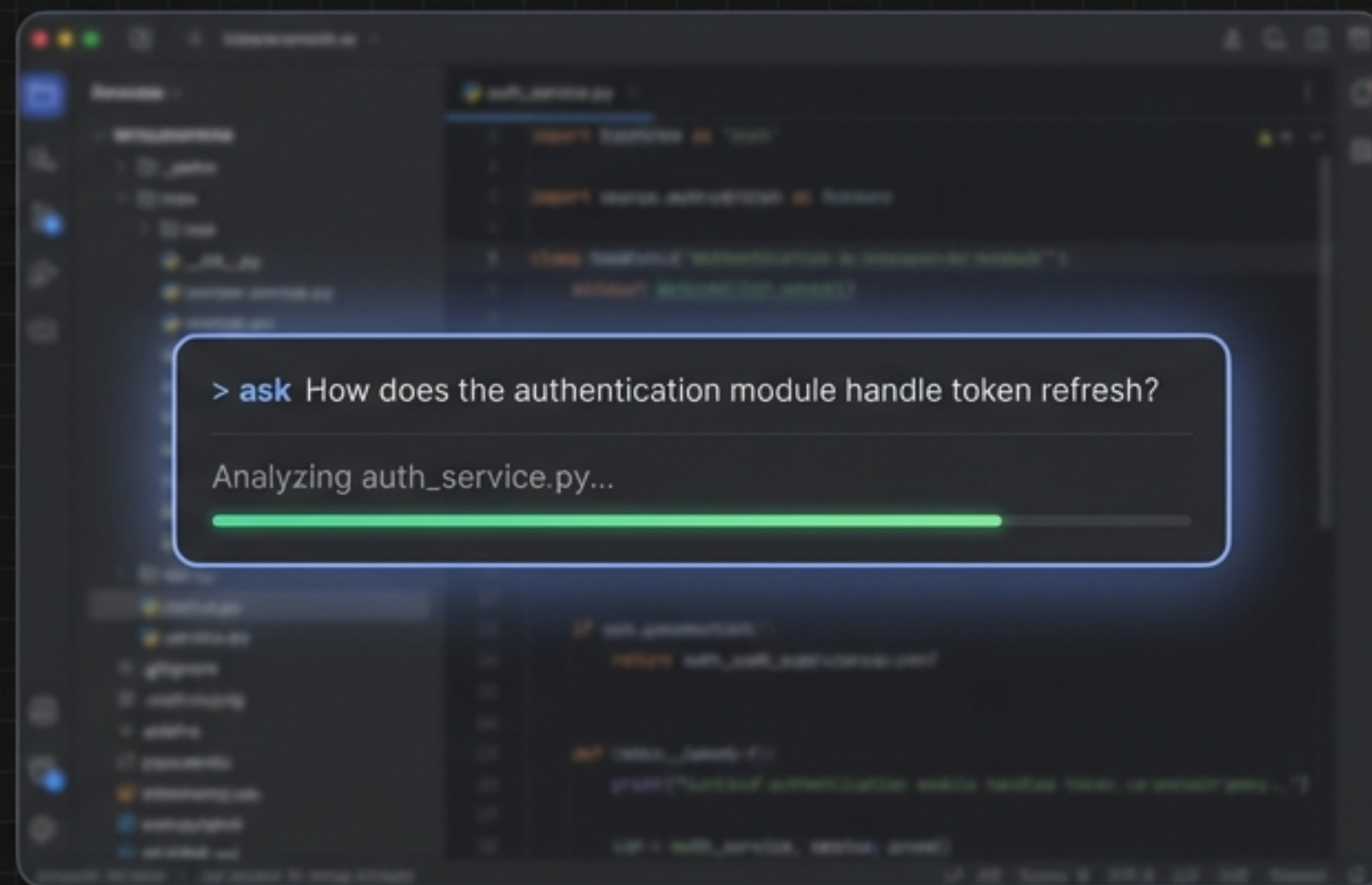
Goal: Maintain flow state by eliminating context switching.

search <query>: Full-text search in <1 second.

ask <question>: Natural language queries about the workspace.

watch: Real-time index updates on file change.

explain <file>: AI-driven module explanation.



"Reading time: 10 min. Solves the problem of breaking focus to find documentation."

Perspective: Architecture & Operations

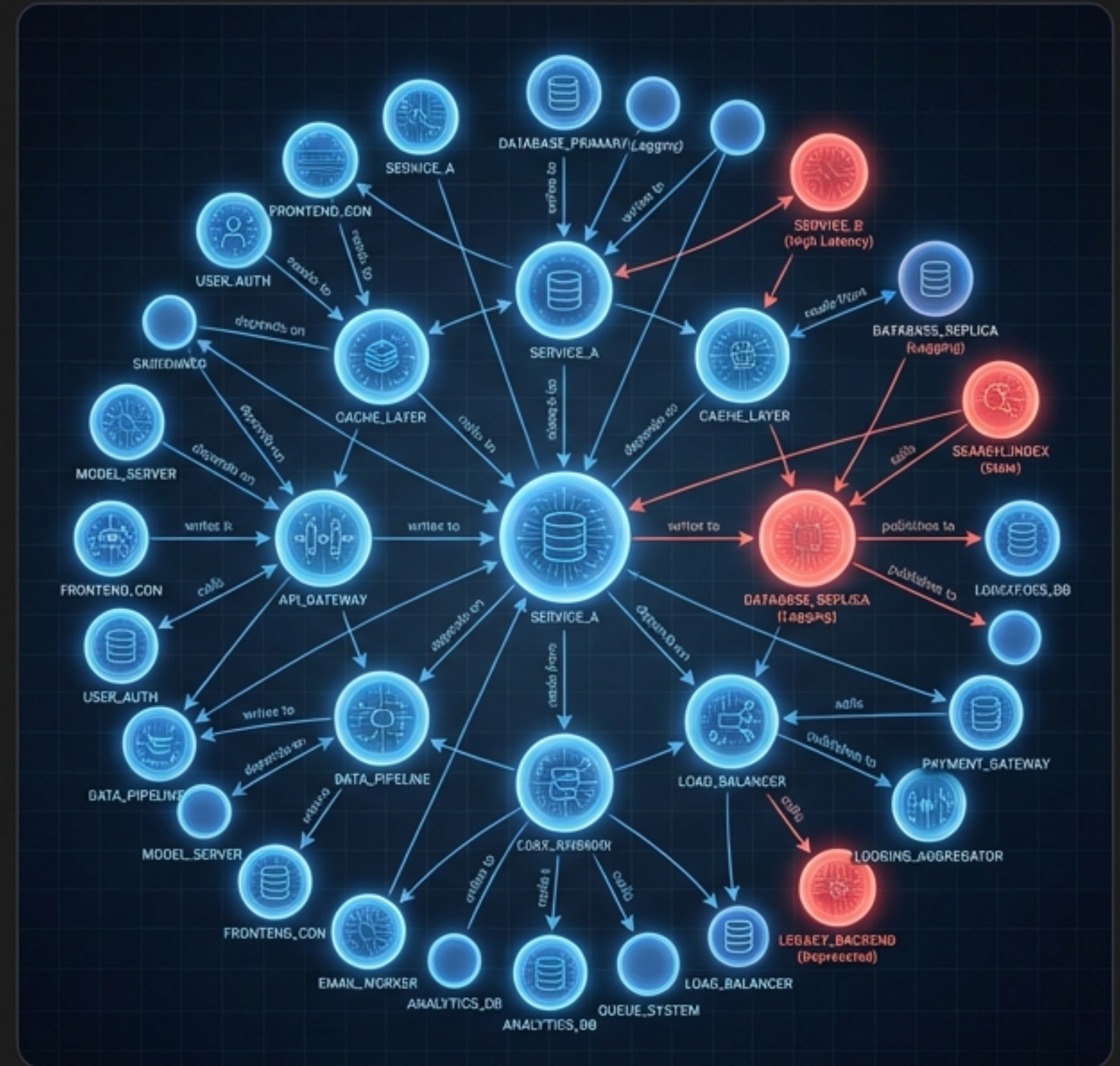
Goal: Visualize system health, dependencies, and deployment structure.

Architect Capabilities

1. **graph**: Generate visual maps (Mermaid, PNG).
2. **architecture**: Detect anti-patterns and coupling.
3. **cross-repo-deps**: Map dependencies across the ecosystem.

DevOps Capabilities

1. **maintain**: Incremental index updates.
2. **credentials**: Secure local key management.



Perspective: The Executive View

The "exo" wrapper abstracts code into business intelligence.

`exo report`: Weekly CEO briefing on velocity.

`exo decisions`: Highlights critical technical choices.

`exo pipeline`: Summary of recent progress.

`exo <name>`: Smart query for Client or Product status.



Reading time: 5 min. Designed for high-level decision making without digging into code.

Perspective: AI Agents & MCP Clients

Synthesis serves as the knowledge layer for autonomous agents.



-  **Programmatic Access:** Agents use `search --all` and `ask` to gather context.
-  **Tool Capabilities:** Grants agents the ability to 'see' the entire codebase structure.
-  **Future-Proofing:** Ready for the age of autonomous coding.

Deep Analysis & Natural Language Understanding

Powered by models like `claude-sonnet` (configurable).

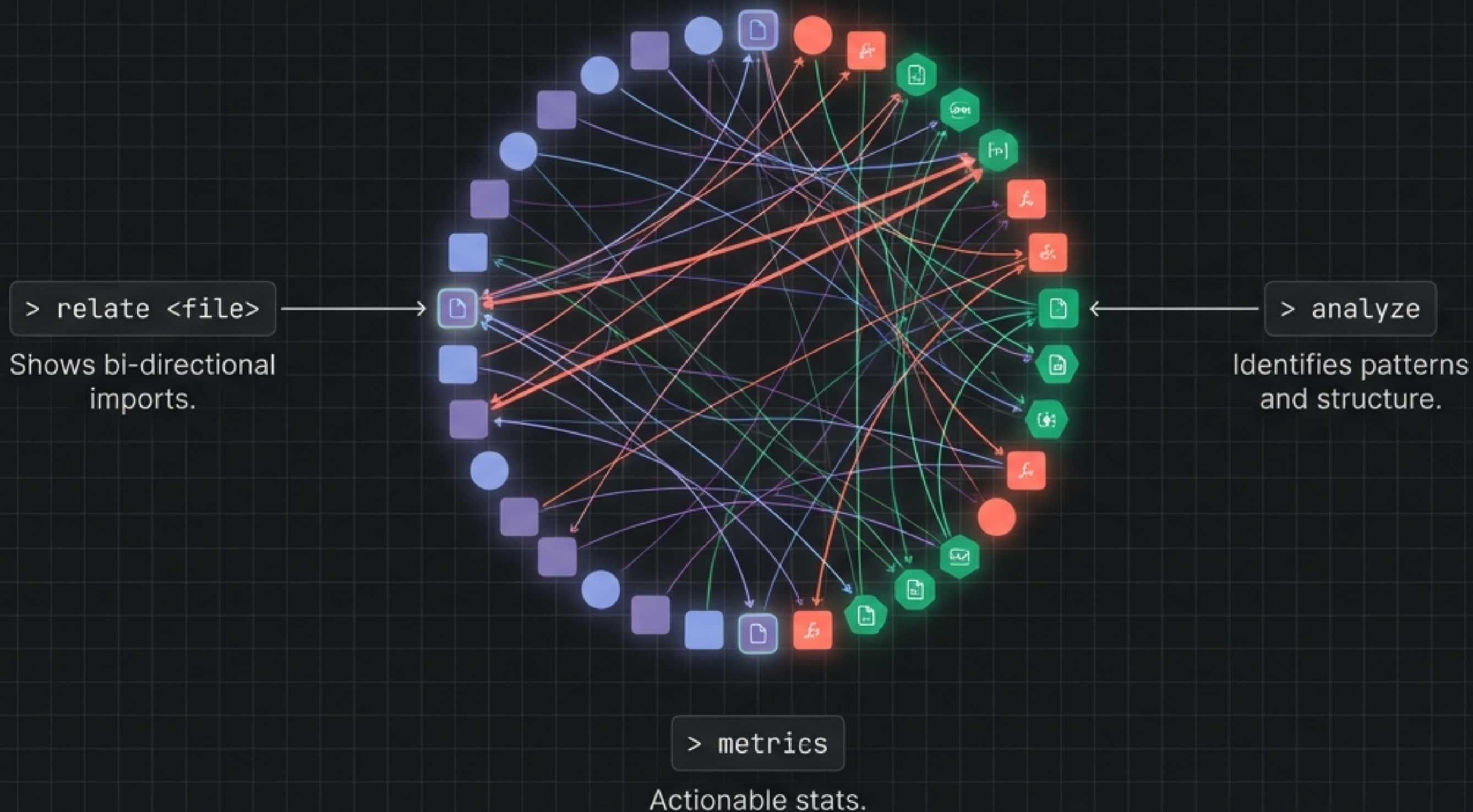
Commands:

- ✂ **perspectives** `<question>`: Analyze complex problems from multiple viewpoints (Security vs Performance).
- **research**: Deep, multi-pass codebase analysis for audits.
- **learn**: Generate skills derived from workspace knowledge.



Structural Intelligence & Graphing

Synthesis understands relationships, not just text.



Beyond Code: Organizational Knowledge



Content Enrichment

Root

ret

Images

videos

Images

architecture

images

walkthrough

videos

.gitignore

.gitignore

README.md

README.md

walkthrough.mp

- enrich** : Generates searchable text for binary assets (Images, Videos, PDFs).
- extract-slides** : Converts PDF decks to indexable images.

../assets/images/architecture.png → OCR Result

../assets/videos/walkthrough.mp4 → Transcription

../assets/videos/walkthrough.mp4 → Transcription

../decks/quarterly_plan.pdf → Images

slide.1.png

slide.2.png

slide.3.png

slide.4.png

slide.2.png

slide.5.png



Organizational Structure

Root

ret

Images

videos

Images

architecture

images

walkthrough

videos

.gitignore

.gitignore

README.md

README.md

walkthrough.mp

- org_scan** : Auto-discovers Companies, Clients, and Products.
- org_classify** : Tags files by business entity.

Companies/

Companies

Companies Clients

Clients/

Clients

Clients

Products/

Products

../contracts/agreement_acme.pdf → Company: Acme Corp

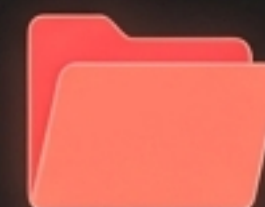
../contracts/agreement_acme.pdf → Client: Acme Corp

../contracts/agreement_acme.pdf → Client: Acme Corp

../contracts/agreement_acme.pdf → Client: Acme Corp

../contracts/agreement_acme.pdf → Client: Acme Corp

../contracts/agreement_acme.pdf → Client: Acme Corp



Exportability

Root

ret

Images

videos

Images

architecture

images

walkthrough

videos

.gitignore

.gitignore

README.md

README.md

export_notes.md

- export** : Dumps index to JSON or Markdown for portability.

export

export_data.json

```
{  
  "name": "stcsk",  
  "value": "it"  
},  
{  
  "state": "string",  
  "acceptor": "architecture..."  
}
```

export_notes.md

```
## Barhdeen  
- Export as Stcsk:files  
- Export note later asocasslathy  
  
## Barhdeen  
....
```


Security, Privacy, & Editions

API keys are stored locally with obfuscation (~/.synthesis/credentials`).

	Core (Air-Gapped)	Pro (Cloud)	Enterprise (Air-Gapped)	Ultimate (Cloud)
AI Features	No ❌	Yes ✅	No ❌	Yes ✅
Telemetry	No ❌	Yes ✅	No ❌	Yes ✅
Daemon	No ❌	No ❌	Yes ✅	Yes ✅
Cloud	No ❌	Yes ✅	No ❌	Yes ✅

Edition set via SYNTHESIS_EDITION environment variable.

Rapid Integration: 5 Commands to Value

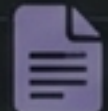
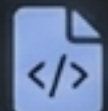
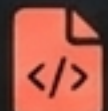
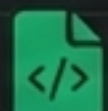
```
1 # 1. Install
2 curl -fsSL ../install.sh | bash
3 # 2. Auth (Optional)
4 synthesis credentials set ANTHROPIC_API_KEY ...
5 # 3. Init
6 synthesis init
7 # 4. Index
8 synthesis scan
9 # 5. Value
10 synthesis search 'authentication'
```

Requirements: Java 17+ (Tested with Java 24).

Transform Your Knowledge Infrastructure

Stop searching. Start synthesizing.

Next Steps

-  EXECUTIVE.md
-  DEVELOPER.md
-  ARCHITECT.md
-  DEVOPS.md

GitHub: github.com/exoreaction/Synthesis
Issues & Support: github.com/exoreaction/Synthesis/issues